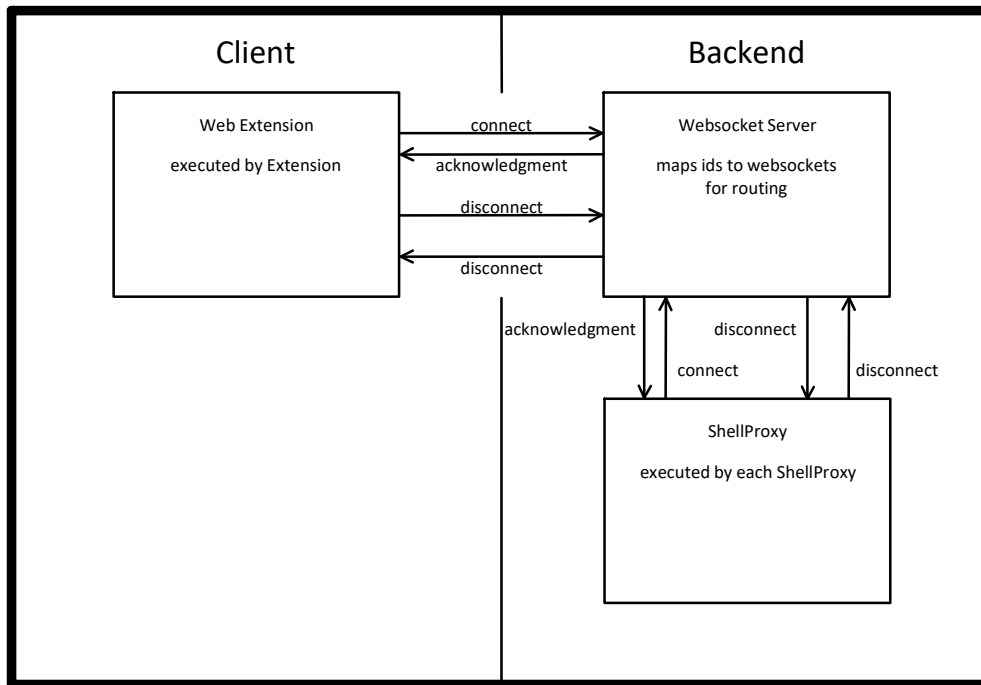


Connect & Disconnect



Connect message

send from Web Extension or ShellProxy to inform Websocket Server of new peer

- **when send from Web Extension to Websocket Server**
 - includes id=0 as unique identifier of Client
- **when send from ShellProxy to Websocket Server**
 - includes id=PID as unique identifier of ShellProxy

Websocket Server maps id to websocket and responds with acknowledgment

Disconnect message

can be send from anywhere to terminate a connection

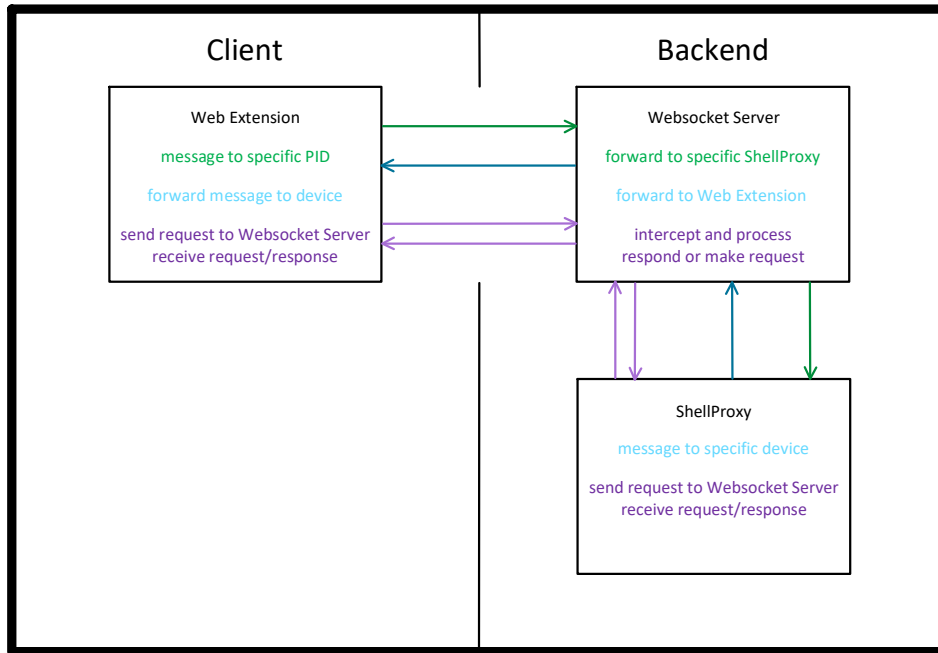
is send from Websocket Server when receiving a non (dis-)connect message from an unknown id
no acknowledgment needed

Connect Message (*->WSS)		
index	content	description
0	"connect"	inform WSS of new connection
1	["client", 0] ["shell", id]	address of peer

Connect Acknowledgment (WSS->*)		
index	content	description
0	"connect ACK"	inform peer of connection success

Disconnect Message (*->*)		
index	content	description
0	"disconnect"	inform peer of connection termination

Websocket Server routing

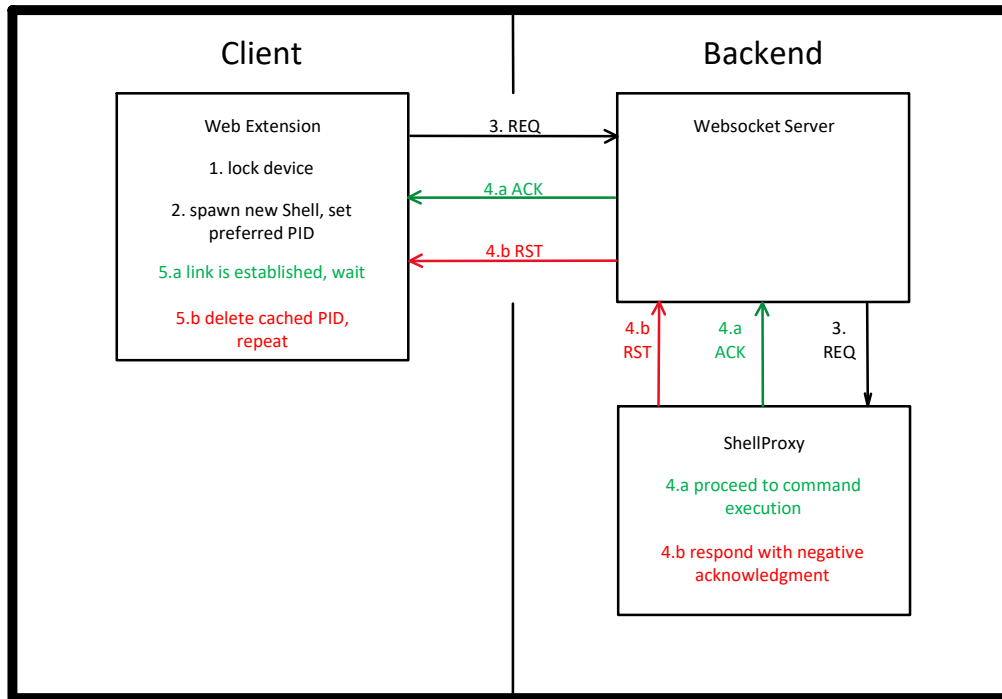


After peers have connected, they are able to send messages

Websocket Server can forward or intercept messages, based on type (index 0)

- **Message from Client to ShellProxy**
 - Websocket Server checks type, decides to forward
 - from=["device", name], to=["shell", id], Websocket Server forwards message to ShellProxy with id
- **Message from ShellProxy to Device**
 - Websocket Server checks type, decides to forward
 - from=["shell", id], to=["device", name], Websocket Server forwards message to Client with id=0
 - (Client has to forward message according to device id)
- **Message intercepted by Websocket Server**
 - Websocket Server checks type, decides to intercept
 - is processed by Websocket Server
 - is able to respond or send own requests

Command Request



Command Request

executed when:

- User presses a Button in the Web Extension

Device has either a cached PID or not

1. Device locks itself
2. (if no cached PID present) spawn new Shell, set cached PID to new shell PID
3. Device sends RM (Request Message) to cached PID
4. ShellProxy responds to Device
 - a. if ShellProxy is in non user mode and idle, respond with acknowledgment, proceed to command execution
 - b. if not, respond with termination
5. Device receives message
 - a. if acknowledgment, link is established, wait for further actions
 - b. else delete cached PID, repeat from step 2

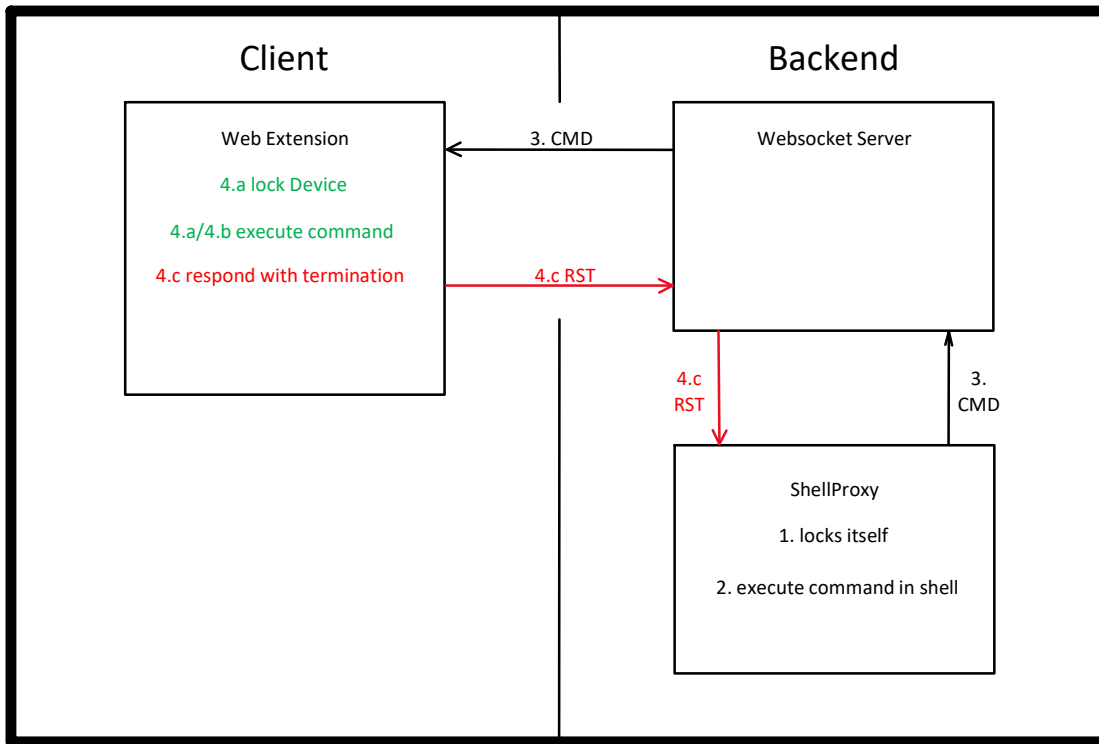
Shell Request Message		
index	content	description
0	"REQ"	inform shell of command
1	["device", name]	address of sender
2	["shell", id]	address of recipient
3	newInstance	true or false, depending on if extension spawned a new shell
4	command request	command that shell should prepare

Request Message Acknowledgment		
index	content	description
0	"ACK"	inform device of successful link establishment
1	["shell", id]	address of sender
2	["device", name]	address of recipient

Command Request		
index	content	description
0	"flash" "term"	type of command
1	board	which board to use
2	projectPath	where to execute command

Termination Message		
index	content	description
0	"RST"	inform device of unsuccessful link establishment
1	["shell", id]	address of sender
2	["device", name]	address of recipient
3	"success" "error"	class of termination
4	msg?	optional message

Command Execution



Command Execution

executed when:

- User presses a Button in the Web Extension (after Request Message)
- User enters command in shell

1. Shell locks itself
2. Shell executes corresponding shell command
3. Shell sends command to Device

Device can have 3 states:

- a. Unlocked (user tries to access unoccupied device)
 - b. Locked to current Shell (after sending a request message)
 - c. Locked to a different Shell (user tries to access occupied device)
4. Device takes action based on state
 - a. locks itself and executes command
 - b. execute command
 - c. respond with termination

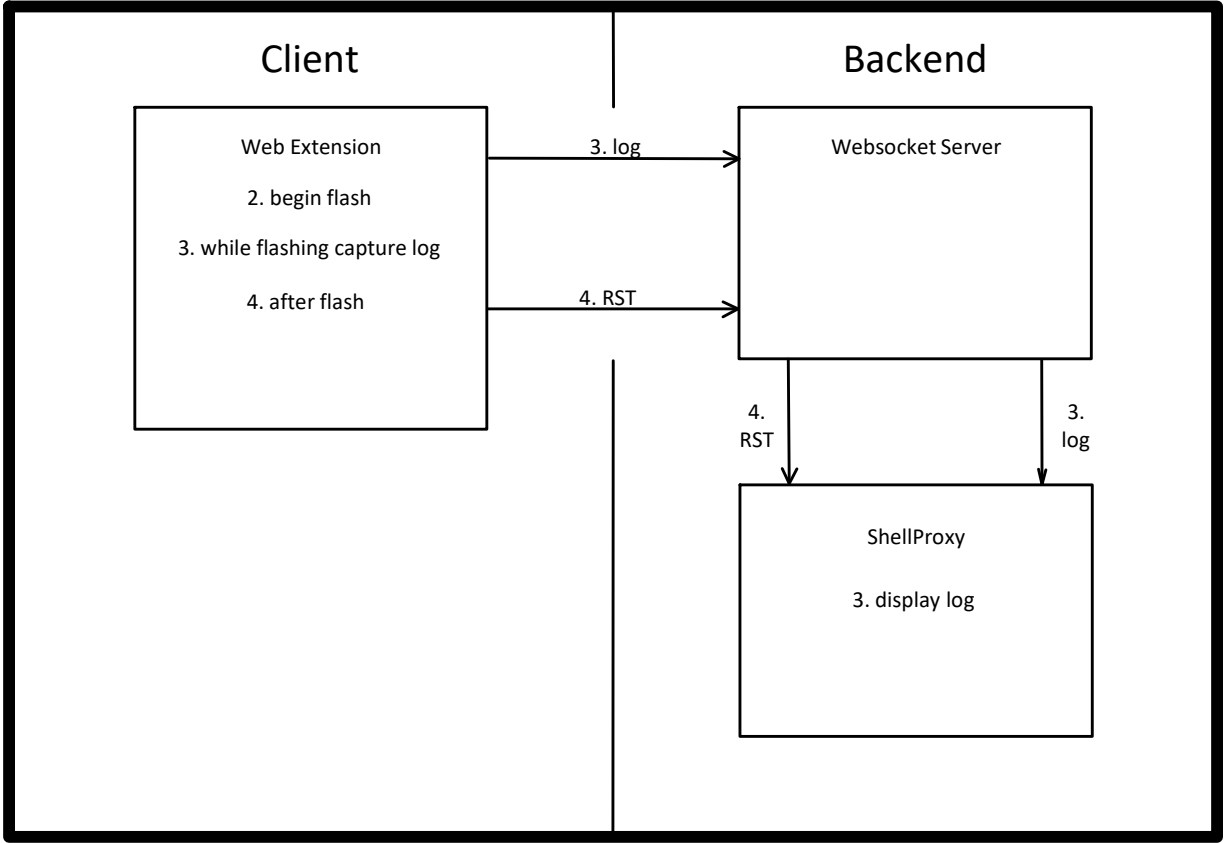
Command Execution Message		
index	content	description
0	"CMD"	execute command on device
1	["shell", id]	address of sender
2	["device", name]	address of recipient
3	flashCommand termCommand	command that device should execute

Flash Command		
index	content	description
0	"flash"	type of command
1	board	used board
2	binaries	binaries to flash (as map from address to binary)
3	arguments	arguments for flashing (as string)

Termination Message		
index	content	description
0	"RST"	inform device of unsuccessful link establishment
1	["shell", id]	address of sender
2	["device", name]	address of recipient
3	"success" "error"	class of termination
4	msg?	optional message

Term Command		
index	content	description
0	"term"	type of command
1	board	used board
2	baudrate	baudrate to use

Flash Command



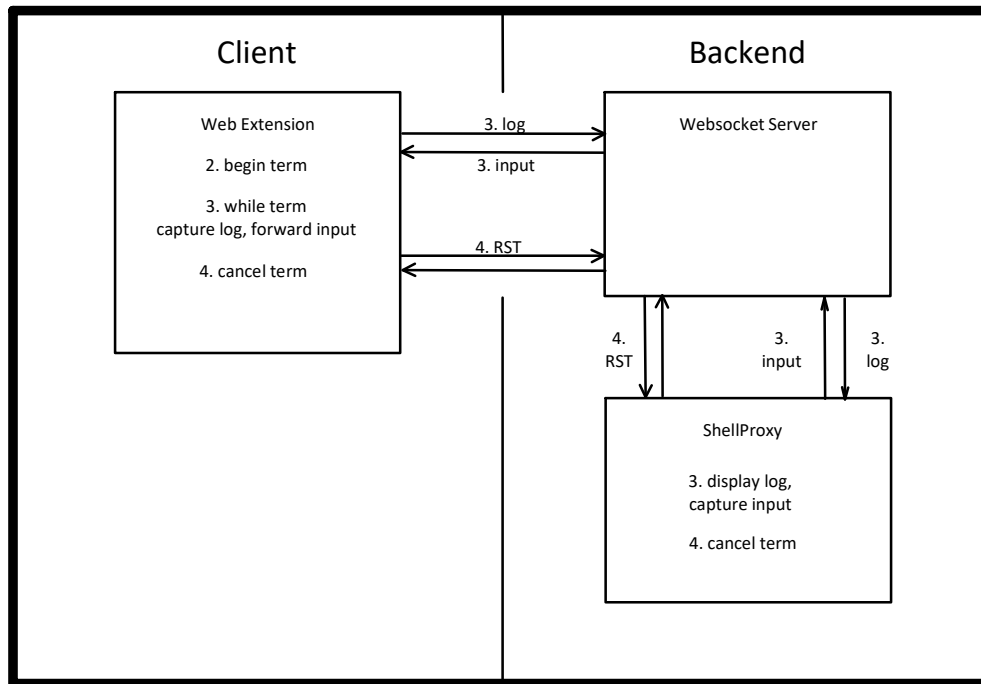
Flash Command

1. device receives flash command
2. device starts flashing
3. device sends log messages to display in ShellProxy
4. On finish, device sends success termination

Flash Log Message		
index	content	description
0	"LOG"	inform shell of log message
1	["device", name]	address of sender
2	["shell", id]	address of recipient
3	"log" "error"	class of log message
4	msg	log message

Termination Message		
index	content	description
0	"RST"	inform device of successful link termination
1	["shell", id]	address of sender
2	["device", name]	address of recipient
3	"success"	class of termination
4	msg?	optional message

Term Command



Term Command

1. device receives term command
2. device starts term
3. device sends log messages to display in ShellProxy, ShellProxy sends input to send to device
4. On cancel (from either side), send link termination

Input/Output Message		
index	content	description
0	"IO"	forward message to device
1	["device", name] ["shell", id]	address of sender
2	["shell", id] ["device", name]	address of recipient
3	msg	input that is forwarded to device

Termination Message		
index	content	description
0	"RST"	inform device of link termination
1	["shell", id]	address of sender
2	["device", name]	address of recipient
3	"success" "error"	class of termination
4	msg?	optional message